

Solutions

DS 120 Fall'25

Time: 50 minutes

Midterm 1 Practice

1. Use truth table to determine whether the following two statements are logically equivalent: $(P \Rightarrow Q) \wedge (P \Rightarrow R)$ and $P \Rightarrow (Q \wedge R)$.

P	Q	R	$P \rightarrow Q$	$P \rightarrow R$	$(P \rightarrow Q) \wedge (P \rightarrow R)$	$Q \wedge R$	$P \rightarrow (Q \wedge R)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	F
T	F	T	F	T	F	F	F
T	F	F	F	F	F	F	F
F	T	T	T	T	T	T	T
F	T	F	T	T	T	F	T
F	F	T	T	T	T	F	T
F	F	F	T	T	T	F	T

Conclusion: For each combination of truth assignments given to the propositional formulas P , Q , and R , we see that the resulting truth value for $(P \rightarrow Q) \wedge (P \rightarrow R)$ and $P \rightarrow (Q \wedge R)$ match.

Therefore they are logically equivalent.

2. Negate the following: $\neg Q \rightarrow P$

(Show proper steps) .

$$\text{Recall that } \neg(A \rightarrow B) \Leftrightarrow A \wedge \neg B$$

$$\text{so, } \neg(\neg Q \rightarrow P) \Leftrightarrow (\neg Q) \wedge \neg P.$$

$$\Leftrightarrow \boxed{\neg Q \wedge \neg P}$$

3. Which of the following matrices are in RREF? 4 points

(Explain why/why not)

$$\begin{array}{cccc} A & B & C & D \\ \begin{bmatrix} 1 & 3 & 5 \\ 2 & 3 & 0 \\ 1 & 0 & 0 \end{bmatrix} & , \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} & , \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} & , \begin{bmatrix} 1 & 0 & 2 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 2 & 0 \end{bmatrix} \end{array}$$

Matrix B is the only matrix in RREF

because:

- All leading entries are one
- Each leading entry is to the right of the one above it.
- All entries above and below leading entries are zero.
- All zeroed-out rows are at the bottom.

Note: leading entries is the same as a pivot / pivot point.

4. Use Gauss Jordan Elimination to solve the following system of equations.

(Show proper steps)

$$\begin{array}{l} x_1 + 5x_3 = -4 \\ x_1 + 4x_2 + 3x_3 = -2 \\ 2x_1 + 7x_2 + x_3 = -2 \end{array} \quad \begin{bmatrix} 1 & 0 & 5 \\ 1 & 4 & 3 \\ 2 & 7 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4 \\ -2 \\ -2 \end{bmatrix}$$

Set up augmented matrix:

$$\begin{bmatrix} 1 & 0 & 5 & | & -4 \\ 1 & 4 & 3 & | & -2 \\ 2 & 7 & 1 & | & -2 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - 2R_1 \end{array} \quad \begin{bmatrix} 1 & 0 & 5 & | & -4 \\ 0 & 4 & -2 & | & 2 \\ 0 & 7 & -9 & | & 6 \end{bmatrix}$$

$$\begin{array}{l} R_2 \rightarrow R_2 \cdot \frac{1}{4} \\ R_3 \rightarrow R_3 - 7R_2 \end{array} \quad \begin{bmatrix} 1 & 0 & 5 & | & -4 \\ 0 & 1 & -1/2 & | & 1/2 \\ 0 & 7 & -9 & | & 6 \end{bmatrix} \quad \begin{array}{l} R_3 \rightarrow R_3 - 7R_2 \end{array} \quad \begin{bmatrix} 1 & 0 & 5 & | & -4 \\ 0 & 1 & -1/2 & | & 1/2 \\ 0 & 0 & -11/2 & | & 5/2 \end{bmatrix}$$

$$\begin{array}{l} R_3 \rightarrow R_3 \cdot \frac{-2}{11} \\ R_2 \rightarrow R_2 + \frac{1}{2}R_3 \\ R_1 \rightarrow R_1 - 5R_3 \end{array} \quad \begin{bmatrix} 1 & 0 & 5 & | & -4 \\ 0 & 1 & -1/2 & | & 1/2 \\ 0 & 0 & 1 & | & -5/11 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 + \frac{1}{2}R_3 \\ R_1 \rightarrow R_1 - 5R_3 \end{array} \quad \begin{bmatrix} 1 & 0 & 0 & | & -19/11 \\ 0 & 1 & 0 & | & 3/11 \\ 0 & 0 & 1 & | & -5/11 \end{bmatrix}$$

$$\boxed{x_1 = -19/11, \quad x_2 = 3/11, \quad x_3 = -5/11}$$

Check: (optional)

$$\begin{bmatrix} 1 & 0 & 5 \\ 1 & 4 & 3 \\ 2 & 7 & 1 \end{bmatrix} \begin{bmatrix} -19/11 \\ 3/11 \\ -5/11 \end{bmatrix} \stackrel{?}{=} \begin{bmatrix} -4 \\ -2 \\ -2 \end{bmatrix}$$

(3x3)

(3x1)

$$\begin{bmatrix} -19/11 + 0 - 25/11 \\ -19/11 + 12/11 - 15/11 \\ -38/11 + 21/11 - 5/11 \end{bmatrix} = \begin{bmatrix} -44/11 \\ -22/11 \\ -22/11 \end{bmatrix} = \begin{bmatrix} -4 \\ -2 \\ -2 \end{bmatrix} \quad \checkmark$$

5. The sum of two numbers is 25. Their difference is 7. Find the numbers.

Solve using Gauss Jordan elimination and show proper steps

$$\begin{cases} x + y = 25 \\ x - y = 7 \end{cases} \Leftrightarrow \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 25 \\ 7 \end{bmatrix}$$

Solve:

$$\begin{bmatrix} 1 & 1 & | & 25 \\ 1 & -1 & | & 7 \end{bmatrix} \quad R_2 \rightarrow R_2 - R_1 \quad \sim \quad \begin{bmatrix} 1 & 1 & | & 25 \\ 0 & -2 & | & -18 \end{bmatrix}$$

$$R_2 \rightarrow R_2 \cdot \frac{-1}{2} \quad \sim \quad \begin{bmatrix} 1 & 1 & | & 25 \\ 0 & 1 & | & 9 \end{bmatrix} \quad R_1 \rightarrow R_1 - R_2 \quad \sim \quad \begin{bmatrix} 1 & 0 & | & 16 \\ 0 & 1 & | & 9 \end{bmatrix}$$

$$\boxed{x = 16, y = 9}$$

Check:

$$\begin{cases} 16 + 9 = 25 \quad \checkmark \\ 16 - 9 = 7 \quad \checkmark \end{cases}$$

6. Prove using mathematical induction that for all $n \geq 1$: **10 points**

$$2 + 5 + 8 + \dots + (3n - 1) = \frac{n \cdot (3n + 1)}{2}$$

(Write each step clearly)

① Base Case: ($n=1$)

LHS: 2 (first term in summation)

$$\text{RHS: } \frac{1 \cdot (3 \cdot 1 + 1)}{2} = \frac{1 \cdot 4}{2} = 2$$

$$\text{LHS} = \text{RHS} \quad \checkmark$$

② Inductive Hypothesis: (Assume for $n=k$)

Assume for some $k \in \mathbb{N}$,

$$2 + 5 + 8 + \dots + (3k - 1) = \frac{k \cdot (3k + 1)}{2} \quad (*)$$

③ Inductive Step: (Prove for $n=k+1$)

Then, show that

$$2 + 5 + \dots + (3k - 1) + (3(k+1) - 1) = \frac{(k+1)(3(k+1) + 1)}{2}$$

holds for $k+1$.

$$\text{LHS: } 2 + 5 + \dots + (3k - 1) + (3k + 2)$$

$$\text{plug in } (*): = \frac{k(3k + 1)}{2} + 3k + 2$$

$$= \frac{3k^2 + k + 6k + 4}{2} = \frac{3k^2 + 7k + 4}{2}$$

$$\text{RHS: } \frac{(k+1)(3(k+1) + 1)}{2} = \frac{(k+1)(3k + 4)}{2} = \frac{3k^2 + 3k + 4k + 4}{2}$$

$$= \frac{3k^2 + 7k + 4}{2}$$

$$\text{LHS} = \text{RHS}.$$

Thus we have shown by the PMI, that

$$2 + 5 + 8 + \dots + (3n-1) = \frac{n(3n+1)}{2}$$

$$\forall n \in \mathbb{N}.$$

7. Determine if the following vectors are orthogonal:

$$\vec{a} = (2, -1, 0)$$

$$\vec{b} = (1, 2, 3)$$

Vectors orthogonal if their dot product is zero.

$$\vec{a} \cdot \vec{b} = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$= 2 - 2 + 0 = \boxed{0} \checkmark$$

8. Multiply the following matrices:

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{bmatrix} \text{ and } \begin{bmatrix} 2 & 0 \\ 1 & 4 \\ 3 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 & 2 \\ 0 & 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 1 & 4 \\ 3 & -2 \end{bmatrix}$$

2×3 3×2

$$= \begin{bmatrix} 2 - 1 + 6 & 0 - 4 - 4 \\ 0 + 3 + 3 & 0 + 12 - 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 7 & -8 \\ 6 & 10 \end{bmatrix}$$

all sides
↗ equal length

↗ 2 sides equal
length

9. Let P be "A triangle is equilateral" and Q be "A triangle is isosceles"

- Is P necessary or sufficient for Q?
- Write your answer as an implies statement

① P is sufficient for Q, but not necessary

② $P \Rightarrow Q$

(an equilateral triangle implies it is isosceles)

10. Given: "For every integer x , if x is even then x^2 is even".

- Write the above statement using mathematical notations.
- Write the negation of the above statement

$$\bullet \forall x \in \mathbb{Z}, \left(\frac{x}{2} \in \mathbb{Z} \Rightarrow \frac{x^2}{2} \in \mathbb{Z} \right)$$

$$\bullet \neg \left(\forall x \in \mathbb{Z}, \left(\frac{x}{2} \in \mathbb{Z} \Rightarrow \frac{x^2}{2} \in \mathbb{Z} \right) \right)$$

$$\Leftrightarrow \exists x \in \mathbb{Z}, \neg \left(\frac{x}{2} \in \mathbb{Z} \Rightarrow \frac{x^2}{2} \in \mathbb{Z} \right)$$

$$\underline{\text{Negation:}} \quad \exists x \in \mathbb{Z}, \left(\frac{x}{2} \in \mathbb{Z} \wedge \frac{x^2}{2} \notin \mathbb{Z} \right)$$

$$\underline{\text{Converse:}} \quad \forall x \in \mathbb{Z}, \left(\frac{x^2}{2} \in \mathbb{Z} \Rightarrow \frac{x}{2} \in \mathbb{Z} \right)$$

$$\underline{\text{Inverse:}} \quad \forall x \in \mathbb{Z}, \left(\frac{x}{2} \notin \mathbb{Z} \Rightarrow \frac{x^2}{2} \notin \mathbb{Z} \right)$$

$$\underline{\text{Contrapositive:}} \quad \forall x \in \mathbb{Z}, \left(\frac{x^2}{2} \notin \mathbb{Z} \Rightarrow \frac{x}{2} \notin \mathbb{Z} \right)$$